

請勿攜去

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二零二五至二六年度上學期科目考試 (期中)

Course Examination 1st Term, 2025-26

科目編號及名稱

Course Code & Title

CSCI4430 Data Communications and Computer Networks

時間

小時

分鐘

Time allowed

0

hours

25

minutes

學號

姓名

Student I.D. No.

Name

This is a **closed-book** exam. One approved calculator is allowed.

There are **3** questions. Answer **ALL** questions in the space provided on the exam paper.

Write down your **student ID** and **name**.

Question	1	2	3	Total
Points	20	20	10	50
Score				

1. (20 points) **Single Choice Questions**

(1) (2 points) Which of the following statements correctly describes the difference between circuit switching and packet switching?

- A. Circuit switching reserves network resources per connection, while packet switching treats each packet independently without prior reservation.
- B. Packet switching has lower transmission delay than circuit switching because it does not require circuit setup.
- C. Circuit switching is more efficient for bursty traffic, while packet switching is better for constant-bit-rate traffic.
- D. Packet switching uses a dedicated path for each connection, while circuit switching shares paths among multiple packets.

Answer: A

(2) (2 points) Which of the following is INCORRECT about performance metrics of a network?

- A. For two server nodes directly connected via a link, the end-to-end delay between them equals the sum of transmission delay and propagation delay.
- B. For two server nodes connected to a switch, the end-to-end delay between them equals the sum of transmission delay, propagation delay and processing delay.
- C. Queueing delay is only related to arrival rate at the queue and transmission rate of the outgoing link.
- D. Persistent overload at a switch will lead to packet drop.

Answer: C

(3) (2 points) Which of the following is INCORRECT about the layers in network stack?

- A. HTTP is an application-layer protocol.
- B. Transport layer should be implemented by both end hosts and the network.
- C. Network layer (L3) routes packets between networks for global delivery.
- D. Layering reduces complexity of the network but also leads to overhead.

Answer: B

(4) (2 points) Which of the following is a key characteristic of the "narrow waist" of the Internet protocol stack (centered on IP)?

- A. IP must be modified whenever new application-layer protocols (e.g., HTTP/3) are introduced.
- B. All application-layer protocols must directly interact with the data link layer to ensure compatibility.
- C. IP decouples upper-layer applications from lower-layer network technologies, enabling independent evolution of both.
- D. The data link layer and physical layer depend on IP to define their frame formats.

Answer: C

(5) (2 points) When a web browser sends an HTTP request, the data is encapsulated as it moves down the protocol stack. At the Transport Layer, what is the payload of a TCP segment?

- A. The IP packet
- B. The Ethernet frame
- C. The HTTP request message
- D. The physical bits

Answer: C

(6) (2 points) What is the primary purpose of the Domain Name System (DNS)?

- A. To encrypt web traffic between clients and servers.
- B. To map human-readable domain names to IP addresses.
- C. To store web page content for faster access.
- D. To manage email delivery between mail servers.

Answer: B

(7) (2 points) What is the primary function of a reverse proxy cache in the context of a Content Delivery Network (CDN)?

- A. To reduce latency for clients by caching frequently accessed content close to them, typically deployed by ISPs.
- B. To reduce the processing and bandwidth load on origin servers by caching content close to the servers themselves.
- C. To translate private IP addresses used within a content provider's network to public IP addresses.
- D. To encrypt all traffic between clients and the origin server for enhanced security.

Answer: B

(8) (2 points) What is the primary motivation for using Clos or fat-tree topologies in modern data center networks?

- A. To minimize the physical number of switches and cables required, thus reducing capital expenditure.
- B. To provide high bisection bandwidth, allowing communication-intensive applications to run without internal network bottlenecks.
- C. To simplify routing by ensuring there is only a single, fixed path between any two servers.
- D. To prioritize traffic going out to the public internet (North-South traffic) over internal server-to-server traffic (East-West traffic).

Answer: B

(9) (2 points) In TCP socket programming, which sequence of system calls is typically used by a server to prepare for and accept a new client connection?

- A. socket() -> connect() -> accept() -> recv()
- B. socket() -> bind() -> listen() -> accept()
- C. socket() -> listen() -> bind() -> send()
- D. socket() -> accept() -> listen() -> bind()

Answer: B

(10) (2 points) In the context of reliable transport protocols, what is the core advantage of Selective Repeat (SR) over Go-Back-N (GBN)?

- A. SR is only suitable for small window sizes, while GBN works for large windows.
- B. SR only retransmits lost packets, while GBN retransmits all unacknowledged packets after a timeout, saving bandwidth in high-error networks.
- C. SR does not need timers, eliminating timer synchronization issues.
- D. SR uses cumulative acknowledgments, which are simpler to implement than GBN's selective acknowledgments.

Answer: B

2. (20 points) **Short Questions**

(1) (10 points) How can we improve the efficiency of HTTP? Briefly explain one approach.

Answer: Persistent connections; pipelining; parallel connections; any other valid thoughts are fine

Notes: Caching or CDN is acceptable

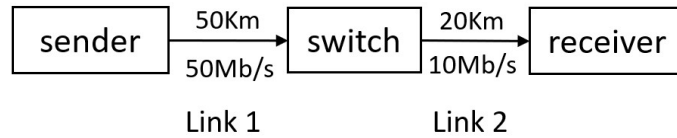
(2) (10 points) Explain the role of the `accept()` system call in TCP socket programming for a server. Why does it create a new file descriptor instead of reusing the listening socket?

Answer:

- After `listen()`: The `accept()` system call is used by a TCP server to handle incoming client connections. It is called after the server calls `listen()` to put the listening socket in a "passive mode" (waiting for connections).
- Block/wait: `accept()` blocks until a client initiates a connection via `connect()`.
- Create a socket: When a connection is established, `accept()` creates a new file descriptor (socket) dedicated to communication with that specific client.
- Different function: It does not reuse the listening socket because the listening socket is reserved for accepting new connection requests (it remains in passive mode to handle other clients).
- New identifier: Using a new socket per client allows the server to distinguish between multiple concurrent connections: each new socket maps to a unique 4-tuple (server IP, server port, client IP, client port), enabling the server to send/receive data to/from each client independently (e.g., via `read()/write()` on the new socket). For example, a web server can use the listening socket (port 80) to accept connections from 100 clients, and `accept()` will create 100 new sockets — one for each client to handle HTTP requests.

3. (10 points) Performance Analysis

Consider the figure below. The sender connects to the receiver through one store-and-forward switch. The sender generates a stream of 1000-byte data packets to the receiver and is assumed to always have data to send. The propagation speed in all links is 2.5×10^8 m/s. The sender, switch, and receiver have negligible processing and queuing delay, and the size of ACK packets is also considered negligible. (Note: 1 Byte = 8 bits)



(a) What is the average throughput of this connection in Mb/s?

Answer: $d_1 = 0.2\text{ms}$, $d_2 = 0.08\text{ms}$, $d_{\text{total}} = d_1 + d_2 = 0.28\text{ms}$

Packet size in bits: $1000\text{bytes} \times 8 = 8000\text{bits}$; $t_1 = 0.16\text{ms}$, $t_2 = 0.8\text{ms}$; $t_{\text{cycle}} = t_1 + t_2 + \text{RTT} = 1.52\text{ms}$;
throughput = $8000 / (1.52 \times 10^{-3}\text{s}) = 5.26 \text{ Mb/s}$

(b) Based on the throughput calculated in part (a), what is the utilization of Link 1 and Link 2?

Answer: $5.26/50 = 10.52\%$; $5.26/10 = 52.6\%$